

main.c

Run

```
1 #include <stdio.h>
2 #define MAX 10
3 int arr[MAX], n;
4 void display()
5 {
6     int i;
7     printf("\nArray Elements: ");
8     for(i=0;i<n;i++)
9         printf("%d ",arr[i]);
10    printf("\n");
11 }
12 void insert()
13 {
14     int pos, item, i;
15     if(n==MAX)
16     {
17         printf("Array is Full\n");
18         return;
19     }
20     printf("Enter position: ");
21     scanf("%d", &pos);
```

Output

Clear

```
Enter number of elements (Max 10): 5
Enter 5 elements:
10
20
30
40
50

Array Elements: 10 20 30 40 50

----- MENU -----
1.Insert
2.Delete
3.Search
4.Display
5.Exit
Enter your choice: 1
Enter position: 3
Enter element: 25
Element Inserted Successfully
```

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Run

```
20 printf("Enter position: ");
21 scanf("%d",&pos);
22 printf("Enter element: ");
23 scanf("%d",&item);
24 if(pos<1 || pos>n+1)
25 {
26     printf("Invalid Position\n");
27     return;
28 }
29 for(i=n;i>=pos;i--)
30     arr[i]=arr[i-1];
31 arr[pos-1]=item;
32 n++;
33 printf("Element Inserted Successfully\n");
34 display();
35 }
36 void delete()
37 {
38     int pos,i;
39     if(n==0)
40     {
41         printf("Array is empty\n");
42         return;
43     }
44     printf("Enter position to delete: ");
45     scanf("%d",&pos);
46     if(pos<1 || pos>n)
47     {
48         printf("Invalid Position\n");
49         return;
50     }
51     for(i=pos;i<n;i++)
52         arr[i-1]=arr[i];
53     n--;
54     printf("Element Deleted Successfully\n");
55 }
```

Output

Clear

40
50

Array Elements: 10 20 30 40 50

----- MENU -----
1.Insert
2.Delete
3.Search
4.Display
5.Exit
Enter your choice: 1
Enter position: 3
Enter element: 25
Element Inserted Successfully

Array Elements: 10 20 25 30 40 50

----- MENU -----
1.Insert
2.Delete

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Run

```
37 - {
38     int pos,i;
39     if(n==0)
40     {
41         printf("Array is Empty\n");
42         return;
43     }
44     printf("Enter position to delete: ");
45     scanf("%d",&pos);
46     if(pos<1 || pos>n)
47     {
48         printf("Invalid Position\n");
49         return;
50     }
51     for(i=pos-1;i<n-1;i++)
52         arr[i]=arr[i+1];
53     n--;
54     printf("Element Deleted Successfully\n");
55     display();
56 }
```

Output

Clear

Enter your choice: 1
Enter position: 3
Enter element: 25
Element Inserted Successfully

Array Elements: 10 20 25 30 40 50

----- MENU -----
1.Insert
2.Delete
3.Search
4.Display
5.Exit
Enter your choice: 2
Enter position to delete: 5
Element Deleted Successfully

Array Elements: 10 20 25 30 50

----- MENU -----
1.Insert

main.c



Run

Output

Clear

```
57 void search()
58 {
59     int key,i,flag=0;
60     printf("Enter element to search: ");
61     scanf("%d",&key);
62     for(i=0;i<n;i++)
63     {
64         if(arr[i]==key)
65         {
66             printf("Element Found at Position %d\n",i+1);
67             flag=1;
68             break;
69         }
70     }
71     if(flag==0)
72         printf("Element Not Found\n");
73 }
74 int main()
75 {
76     int i,ch;
```

```
Element Deleted Successfully

Array Elements: 10 20 25 30 50

----- MENU -----
1.Insert
2.Delete
3.Search
4.Display
5.Exit
Enter your choice: 3
Enter element to search: 25
Element Found at Position 3

----- MENU -----
1.Insert
2.Delete
3.Search
4.Display
5.Exit
Enter your choice: 4
```

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main.c

Run

```
76 int i,ch;
77 printf("Enter number of elements (Max 10): ");
78 scanf("%d",&n);
79 printf("Enter %d elements:\n",n);
80 for(i=0;i<n;i++)
81     scanf("%d",&arr[i]);
82 display();
83 do
84 {
85     printf("\n----- MENU ----- \n");
86     printf("1.Insert\n");
87     printf("2.Delete\n");
88     printf("3.Search\n");
89     printf("4.Display\n");
90     printf("5.Exit\n");
91     printf("Enter your choice: ");
92     scanf("%d",&ch);
93     switch(ch)
94     {
95         case 1: insert(); break;
```

Output

Clear

```
3.Search
4.Display
5.Exit
Enter your choice: 3
Enter element to search: 25
Element Found at Position 3

----- MENU -----
1.Insert
2.Delete
3.Search
4.Display
5.Exit
Enter your choice: 4

Array Elements: 10 20 25 30 50

----- MENU -----
1.Insert
2.Delete
3.Search
```

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main.c

Run

```
85 printf("\n----- MENU ----- \n");
86 printf("1.Insert\n");
87 printf("2.Delete\n");
88 printf("3.Search\n");
89 printf("4.Display\n");
90 printf("5.Exit\n");
91 printf("Enter your choice: ");
92 scanf("%d",&ch);
93 switch(ch)
94 {
95     case 1: insert(); break;
96     case 2: delete(); break;
97     case 3: search(); break;
98     case 4: display(); break;
99     case 5: printf("Program Ended\n"); break;
100    default: printf("Invalid Choice\n");
101 }
102 }while(ch!=5);
103 return 0;
104 }
```

Output

Clear

```
1.Insert
2.Delete
3.Search
4.Display
5.Exit
Enter your choice: 4

Array Elements: 10 20 25 30 50

----- MENU -----
1.Insert
2.Delete
3.Search
4.Display
5.Exit
Enter your choice: 5
Program Ended

=== Code Execution Successful ===
```